Lec-8

31/10/26 11:44 AM

→ 5 6 7 1 2 3 4

6 7 1 2 3 4 5

140 → n

14

1 2 3 4 5 6 7 → N = 7, K = 142

K=1 7 1 2 3 4 5 6

K=2 6 7 1 2 3 4 5

K=3 5 6 7 1 2 3 4

K=4 4 5 6 7 1 2 3

K=5 3 4 5 6 7 1 2

K=6 2 3 4 5 6 7 1

K=7 1 2 3 4 5 6 7 K=7

K=8 7 1 2 3 4 5 6

K=9 6 7 1 2 3 4 5

K=10 5 6 7 1 2 3 4

K = = N →

P.N

① K = K % N

K < N

Same

| 1 | 2 | 3 | 4 | 5 | 6 | 7 |

| 5 | 6 | 7 | 1 | 2 | 3 | 4 |

0 1 2 3 4 5 6

| 1 | 2 | 3 | 4 | 5 | 6 | 7 |

0 1 2 3 4 5 6

| 7 | 1 | 2 | 3 | 4 | 5 | 6 |

0 1 2 3 4 5 6

① 7 ← item = arr(6) ↳ N-1

x

| 7 | 1 | 2 | 3 | 4 | 5 | 6 |

0 1 2 3 4 5 6

N =

i = N-2

⑪ for(i = n-2 i>=0 i--){
arr[i+1] = arr[i]

}

i=5 arr[6] = arr[5]

i=4 arr[5] = arr[4]

i=3 arr[4] = arr[3]

i=2 arr[3] = arr[2]

i=1 arr[2] = arr[1]

i=0 arr[1] = arr[0]

⑭ i=0 arr[0] = item

① Reversal Algorith

K = 3

n

n-k-1, n-k-1, K

0

| 1 | 2 | 3 | 4 | 5 | 6 | 7 |

0 1 2 3

4 3 2 1 5 6 7

n-k-1, n-k, n-1

| 1 | 2 | 3 | 4 | 7 | 6 | 5 |

0 1 2 3 4 5 6

5 6 7 4 3 2 1

① 1st, Start n-k
K

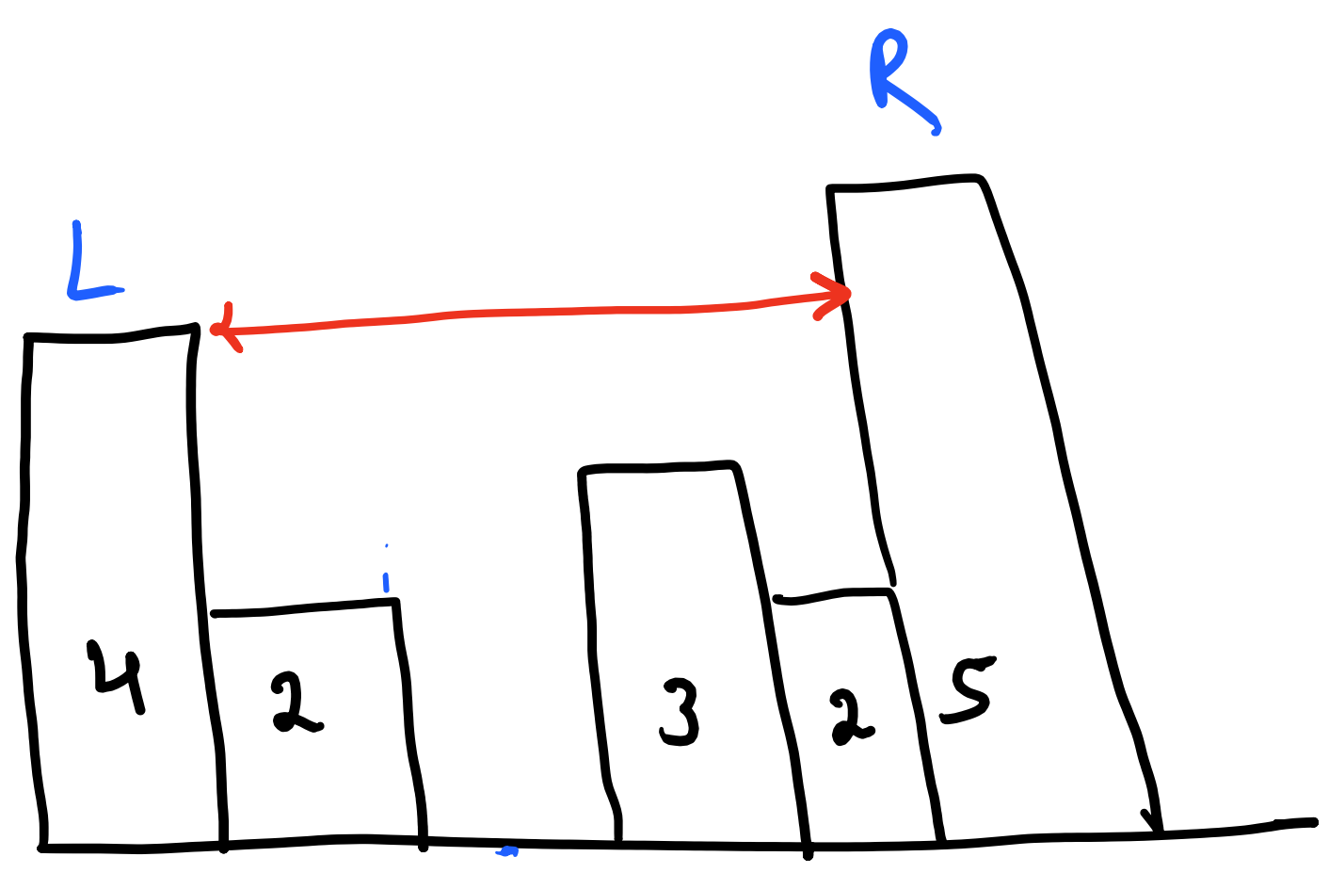
i = 0

j = n-k-1

| 1 | 2 | 3 | 4 | 5 | 6 | 7 |

7 6 5

```
height = [4,2,0,3,2,5]
```



i → [0 - i] max
(i+1) n-1 max

[5, 3, 1, 2, 7, 4, 1, 6]
0 1 2 3 4 5 6 7

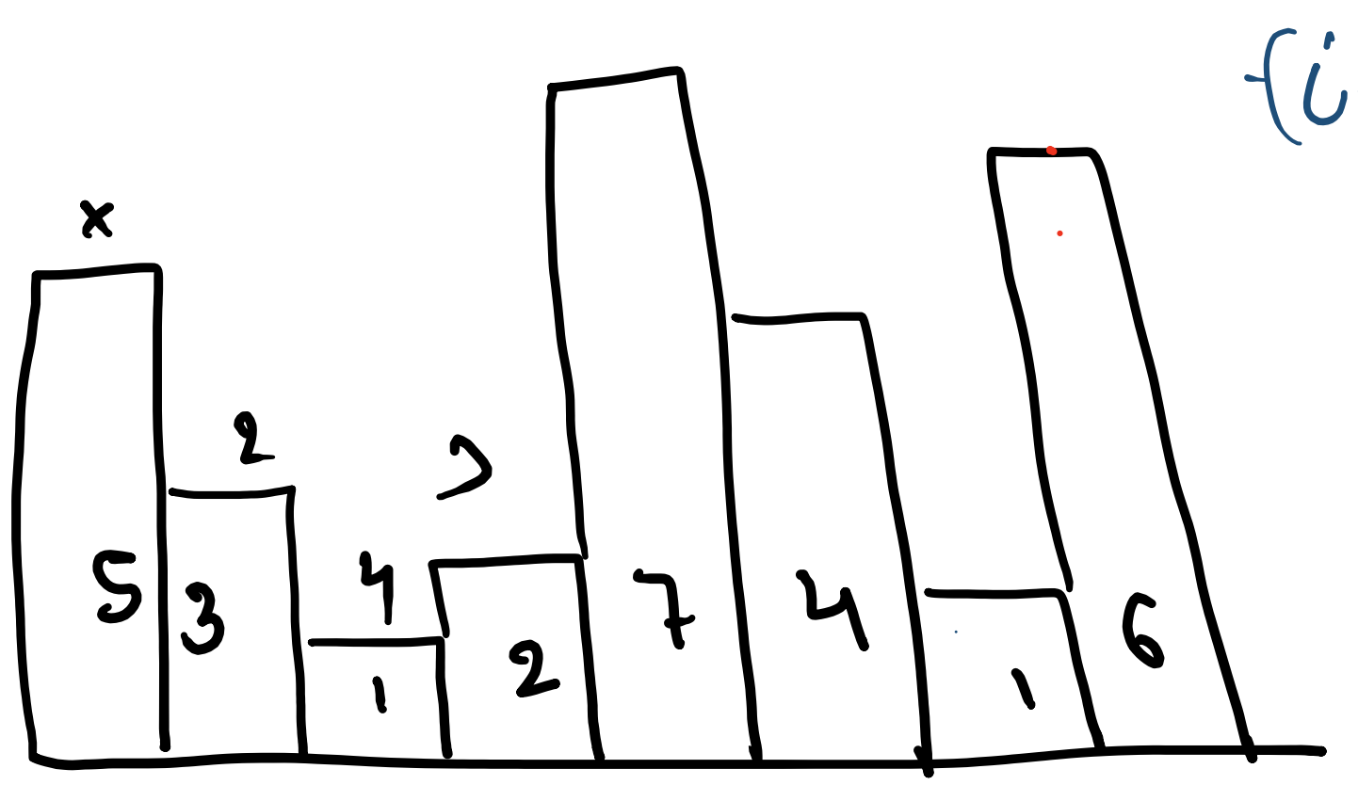
| 5 | 5 | 5 | 5 | 7 | 7 | 7 | 7 |

0 1 2 3 4 5 6 7

①

| 7 | 7 | 7 | 7 | 7 | 6 | 6 | 6 |

0 1 2 3 4 5 6 7



i=0 min(L[0], R[0]) - arr(0) = (5,7) - 5 = 0

i=1 min[L[1], R[1]] - arr[1] = (5,7) - 3 = 2

i=2 min(L[2], R[2]) - arr(2) = (5,7) - 1 = 4

i=3 min (L[3], R[3]) - arr(4) = (5,7) - 2 = 3

i=4 min (L[4], R[4]) - arr(4) = (7,7) - 7 = 0

i=5 min (L[5], R[5]) - arr(5) = (7,6) - 4 = 2

i=6 min (L[6], R[6]) - arr(6) = (7,6) - 1 = 5

i=7 min [L[7], R[7]) - arr(7) = (7,6) - 6 = 0

for(i=0, i<n i++){
sum += (min[L(i), R(i)) - arr(i))
}

L(i) = max(L(i-1), arr(i))

L(1) = max(L(0), arr(1))

L(2) = max(L(1), arr(2))

L(3) = max(L(2), arr(3))

```
height = [4,2,0,3,2,5]
```

0 1 2 3 4 5

| 4 | 4 | 4 | 4 | | |

0 1 2 3 4 5

| | | 5 | 5 | 5 | 5 |

0 1 2 3 4 5

R(n-1) = arr(n-1)

R(4) = max(R(5), arr(4))

R(3) = max(R(4), arr(3))

R(2) = max(R(3), arr(2))